

Semantic Web and Information Extraction Technologies

Semantic Web, Fall 2026

Preliminary:

- Recommended:
 - download Protégé 5.6.9 at <https://protege.stanford.edu/software/> for either Windows or other OS (this is a 120Mb file), see also <https://github.com/protegeproject/protege-distribution/releases>
 - unzip the archive in Home (i.e., in your /homes/)
 - on Windows: simply run 'Protégé.exe'
 - on other OSs: open a terminal and enter 'cd' to be in your /homes/ and 'run.sh &'
 - it requires a Java Runtime Environment, <https://www.java.com/en/download/manual.jsp>
- Documentation: <http://protegewiki.stanford.edu/wiki/Protege4UserDocs>

Warning: save frequently your work (Protégé can crash) and handle the reasoner carefully!

Instructions:

- Read all questions before starting! Follow strictly **ALL instructions**
- In this assignment, you will need to produce a **report** and an **ontology**
 - Save your project, export the ontology in OWL. The file name SHOULD be: yourlastName-cycling.owl
 - The report should be a file named yourlastName-cycling.pdf or .doc and should include:
 - Yours complete name;
 - The answers to the questions 1), 2), 5), 6), 7), 9), 11) and 12);
 - The questions you haven't been able to do;
 - The reasons why you have found difficult to answer these questions.
 - Upload your ontology and your report on Moodle before **Friday 2 October 2026, 23:59 CET**.

Modeling your first ontology with Protégé: The Cycling domain (20 pts)

1. Start Protégé and imagine a good namespace prefix and a namespace URI for modeling the Cycling domain. Provide this couple prefix/uri in your report and justify why you choose them. (0.5 pts)
2. In the cycling domain, a Race is either a OneDayRace or a SeveralStagesRace. There are **no** other types of race. Model these three classes in the ontology and write in the report the logical formula equivalent to this definition. (1,5 pts)
3. A SeveralStagesRace is composed of Stages. Add the corresponding class using the relationship `composedOf` that you need to create (1 pt)
4. A Stage can be of different type, such as MountainStage, FlatStage, TimeTrial but this list is **not** exhaustive. Further, a TimeTrial is either a IndividualTimeTrial or a TeamTimeTrial. Add all the necessary classes and axioms. (2 pts)
5. Define with necessary and sufficient conditions (i.e. strict equality) the Prologue class such as the Prologue is exactly always the first Stage of a SeveralStagesRace. You might have to introduce more properties or use the inverse of an existing property to do this

- modeling. Write in the report the logical formula equivalent to this definition. Hint: in Protégé, you can use the keyword `value`. (2 pts)
6. Add the concept `Person` to the ontology. A `Person` can be a `Spectator`, a `RacePerson` or a `TeamPerson` among others. A `TeamPerson` is **exactly** the union of a `Doctor`, a `Director` or a `RaceCyclist`. A `RaceCyclist` can himself be considered as a `Rider`, a `Sprinter` or a `Climber`. A `Person` has a name, an age, and a nationality. A `Person` participates in `Race`. Add all these classes and properties and the corresponding axioms. Explain in the report what are the differences between object and datatype properties. (4 pts)
 7. A `Team` is exactly composed of 1 `Doctor`, 1 `Director` and 10 `RaceCyclist`. Define this class using the `belongsTo` property. Explain in the report what can you say about this new property with respect to previously defined properties? (2 pts)
 8. A `GoodTeam` is a `Team` in which there are exactly 1 `Rider` and at least 3 `Climber`. Add this class to your ontology. (1 pt)
 9. Load (import) the FOAF ontology using its <http://xmlns.com/foaf/spec/> and align some of the concepts you have defined such as: the class `Person` is equivalent to the class `Person` defined in FOAF and the class `Team` is a subclass of the class `Organisation` defined in FOAF. Search for properties in the FOAF ontology to align with your own properties and add the corresponding mapping axioms. Write in the report the mappings you have discovered. (2 pts)
 10. Add some instances of the classes: `Rider`, `Climber`, `Team`, `SeveralStagesRace` with e.g.: Tour de France, Vuelta, Giro, Eddy Merckx, Miguel Indurain, Christopher Froome, Astana, Sky, US Postal. (1 pt)
 11. Use the LOV search function at <https://lov.linkeddata.es/dataset/lov/> and identify relevant classes and properties that could be re-used in your ontology. Write them down in your report. (2 pts)
 12. Validate your ontology by clicking on `Reasoner` and `Start reasoner` where `HermiT` is selected as the reasoning service. What is the result? Did you get the message ***“your ontology is coherent and consistent”***? (1 pt)